Overcrowding stress decreases macrophage activity and increases Salmonella Enteritidis invasion in broiler chickens.

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Overcrowding stress is a reality in the poultry industry. Chickens exposed to long-term stressful situations present a reduction of welfare and immunosuppression. We designed this experiment to analyse the effects from overcrowding stress of 16 birds/m(2) on performance parameters, serum corticosterone levels, the relative weight of the bursa of Fabricius, plasma IgA and IgG levels, intestinal integrity, macrophage activity and experimental Salmonella Enteritidis invasion. The results of this study indicate that overcrowding stress decreased performance parameters, induced enteritis and decreased macrophage activity and the relative bursa weight in broiler chickens. When the chickens were similarly stressed and infected with Salmonella Enteritidis, there was an increase in feed conversion and a decrease in plasma IgG levels in the stressed and Salmonella-infected birds. We observed moderate enteritis throughout the duodenum of chickens stressed and infected with Salmonella. The overcrowding stress decreased the macrophage phagocytosis intensity and increased Salmonella Enteritidis counts in the livers of birds challenged with the pathogenic bacterium. Overcrowding stress via the hypothalamic-pituitary-adrenal axis that is associated with an increase in corticosterone and enteritis might influence the quality of the intestinal immune barrier and the integrity of the small intestine. This effect allowed pathogenic bacteria to migrate through the intestinal mucosa, resulting in inflammatory infiltration and decreased nutrient absorption. The data strengthen the hypothesis that control of the welfare of chickens and avoidance of stress from overcrowding in poultry production are relevant factors for the maintenance of intestinal integrity, performance and decreased susceptibility to Salmonella infection.